Looking at cheek cells

A2
Core



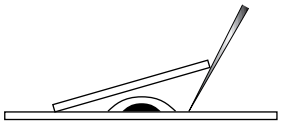
Aim

To make drawings of cheek cells using the microscope.

Equipment

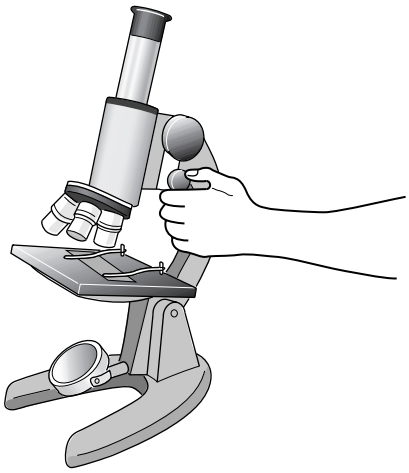
- microscope
- microscope slides
- cover slips
- mounted needle
- cotton buds
- methylene blue
- disinfectant





What to do

1. Wipe a clean fresh cotton bud around the inside surface of your cheek and the outer lower side of the gum.
2. Dab the cotton bud on the centre of a microscope slide, add one drop of methylene blue and cover with a cover slip. The picture shows you how to do this properly.
3. Put the slide on the microscope stage and focus the microscope on a group of cells under low power.
4. Rotate the objective lens to medium power and re-focus using the fine focusing knob.



5. After use, ensure you put the cotton bud and the microscope slide directly into the disinfectant.

Results

Draw what you can see on a blank piece of paper. Look at Topic A2 for hints on how to do really good drawings. Make sure you give your drawing a heading and calculate the total magnification:

total magnification = objective lens magnification × eyepiece lens magnification

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Looking at other types of cell

A2
Extension



Aim

To look at different types of cell under the microscope.

Equipment

- microscope
- mounted needle
- microscope slides
- cover slips
- iodine solution
- scalpel
- onion
- red petals
- plant roots
- potatoes
- tomatoes
- ruler

What to do

1. Peel away the dead, dry layers from the outside of an onion. Break a layer off, and peel off the thin skin (epidermis) on its inside surface.
2. If you find this difficult, crack the layer down the middle, and the skin may peel off between the two halves.
3. Cut or tear your piece of epidermis so it is no bigger than 7 mm square. Always cut with the scalpel facing away from you, and use a mat or a cutting tile.
4. Place it in the centre of a microscope slide, making sure it is flat and not folded.
5. Add one drop of iodine and cover with a cover slip.
6. Put the slide on the microscope stage and focus the microscope on a group of cells under low power.
7. Rotate the objective lens to medium power and re-focus using the fine focusing knob.

Results

Draw and label a group of five cells under medium power.

Extension

1 You have seen how to make slides of scraped cells (cheek cells) and layers of cells (onion cells). Make new slides of petal cells, root cells, potato cells and tomato cells. Draw what you see under medium power.

2 By putting a ruler under the objective lens, measure the width of the field of view. By estimating how far across the field of view each specimen stretches, estimate the size of each specimen you have prepared.

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How to use the microscope

A2
Help



Aim

To learn how to use the microscope.

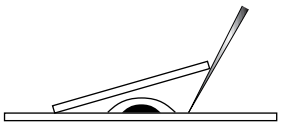
Equipment

- microscope
- cover slips
- mounted needle
- microscope slides
- sand
- newsprint

What to do

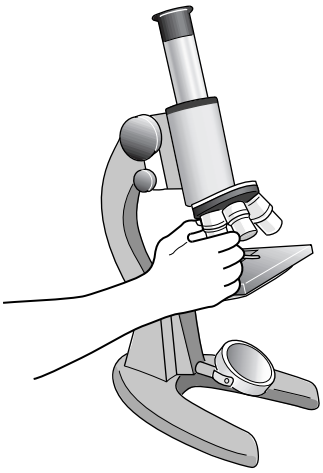
Preparing a slide

1. Cut out a letter ‘‘e’’ from the newspaper you have been given.
2. Place it in the centre of a microscope slide.
3. Put one drop of water on top of the letter.
4. Lower a cover slip onto the drop. The picture shows you how to do this properly.



Looking at a slide

1. Put the slide on the stage, and move it under the stage clips.
2. Check that the ‘‘e’’ is above the centre of the hole in the stage.
3. Rotate the objective lenses until the shortest lens is pointing downwards as shown on the right.
4. Twist the focusing knob until the stage and objective lens are as close as they can be.
5. Look down the microscope and twist the focusing knob towards you.
6. Stop when you can see the ‘‘e’’ clearly and sharply.



Increasing the magnification

1. Twist the objective lenses until the middle lens is pointing downwards.
2. Look down the microscope and twist the fine focusing knob until you can see part of the ‘‘e’’ clearly and sharply.

Results

1 Draw what you see under the microscope on low power and medium power.

2 Use the same method to look at sand and hair under the microscope.

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Making models of animal and plant cells

A3
Core



Aim

To make models of an animal cell and a plant cell.

Equipment

- cellulose paste
- black marker pen
- green marker pen
- two ping-pong balls
- polystyrene
- two large clear plastic bags
- one small clear plastic bag
- cotton or wire to be used for sealing bags
- clear plastic box

What to do

Animal cell

1. Colour a ping-pong ball black and put it into the large plastic bag.
2. Make up the cellulose paste, pour it into the bag, and seal it.

Plant cell

1. Fill the small plastic bag with water, seal it, and put it into the large plastic bag.
2. Roll ten pieces of the polystyrene into small balls (smaller than the ping-pong ball) and colour them green.
3. Colour the ping-pong ball black.
4. Put the polystyrene and the ping-pong ball into the large plastic bag.
5. Make up the cellulose paste, pour it into the bag, and seal it.
6. Place the bag in the plastic box and close the lid.

Question

Each part of the model represents a particular part of the cell. Copy and complete the table using the words below to show how they match up.

cytoplasm chloroplasts nucleus cell membrane cell wall vacuole

Part of the model	Part of the cell
polystyrene balls	
plastic box	
small bag of water	
cellulose paste	
large plastic bag	
ping-pong ball	

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What do pollen grains look like?

A6
Core



Aim

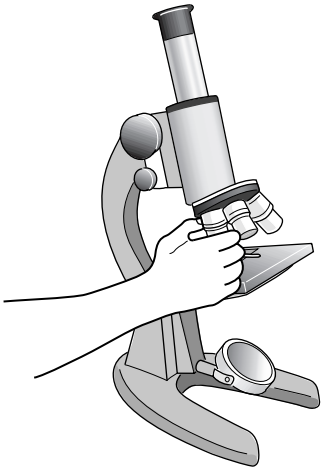
To examine the surface of pollen grains.

Equipment

- microscope
- microscope slide
- cover slips
- mounted needle
- teat pipette
- scissors
- paint brush
- flower with anthers bearing pollen

What to do

1. Remove one anther from your flower using the scissors.
2. Place it on your microscope slide and put it onto the microscope stage.
3. Look at the anthers under low power. Try to find the tiny round pollen grains clinging to their surface. Avoid contact with pollen, especially if you already suffer from hay fever.
4. Remove the slide from the microscope and tap the anthers onto the slide. Some pollen should drop onto the slide. If not, try using the paint brush to transfer the pollen from the anther to the slide.
5. Add one drop of water and cover with a cover slip.
6. Replace the slide on the microscope, and focus on the pollen grains under low power.
7. Twist the objective lens to medium power and focus again.



Results

Draw the pollen grains that you can see. You may need to look on high power to see what the pollen grains look like on their surface.

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How does sugar concentration affect pollen tube growth?



Aim

To find out whether sugar concentration affects how many pollen tubes germinate.

Equipment

- microscope
- cavity microscope slides
- cover slips
- mounted needle
- teat pipette
- scissors
- paint brush
- flower with anthers bearing pollen
- sucrose solutions (10%, 8%, 6%, 4%, 2%, 0%)

Plan and predict

1. Write a plan for the investigation using some or all of the equipment above. When you think about your method, you need to consider:
   - The other variables you need to control (e.g. temperature)
   - How many pollen grains you need to look at to give you a reliable result
   - How you are going to record your results
2. Predict what you think the result will be. Before you begin any practical work, your plan must be checked by a teacher.

Analyse

1. Plot a graph of your results with concentration along the *x*-axis (the one which goes across) and number of germinated pollen grains on the *y*-axis (the one which goes up). Draw a line through the points.
2. As sugar concentration increases, does the number of germinating pollen grains increase?

Evaluate

Compare your graph to those obtained by other groups. Are your graphs identical? If they are not, try to explain why. Answering the following questions will help get you started.

- Are all pollen grains identical?
- Were all anthers from the same flower?
- Was it easy to see if pollen grains had germinated?
- Did any results seem out of place?

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